DEEPFAKE DETECTION

A comparative analysis on custom curated 12K-image deepfake dataset

Anandvardhan Singh Chauhan(E23SEU1246)
Deepankar Shokeen (E23SEU1257)
Vardhan Shandilya(E23SEU1236)
Mahak Taneja (E23CSEU1253)

Introduction

- Deepfakes are synthetic visual media created using deep learning.
- They pose major challenges in misinformation, fraud, and privacy risks.
- Objective: build a lightweight deepfake detector trained on a smaller curated dataset.
- Compare two models: STIL (simple image-level CNN) vs F3-Net baseline (feature-forensics network).

Problem Statement

- Full WildDeepfake dataset is huge (~400K frames).
- Cannot train heavy models directly due to resource limitations.
- Need:
- A balanced small dataset (real vs fake)
- Trainable on Colab GPU
- Good generalization performance
- Evaluate different architectures to find best accuracy on limited data.

Dataset: WildDeepfake

- **Source: https://huggingface.co/datasets/xingjunm/WildDeepfake**
- **Contains:**
  - **700+ real videos**
  - **700+ fake videos**
  - **Millions of extracted frames**
- **Real-world distribution:**
  - **Different cameras**
  - **Various lighting**
  - **High diversity of manipulation methods**
- **Highly challenging and realistic dataset.**

Custom Subset Creation

To fit our training constraints, we created a 12,000-image subset:

- **Train:**
  - **5000 fake**
  - **5000 real**
- **Val:**
  - **1000 fake**
  - **1000 real**
- **Extracted using:**
  - **HuggingFace FileSystem API**
  - **WebDataset streaming**
  - **MTCNN-based face cropping**

```
========================================
Finished.
Shards tried: 82
Seen samples: 65200
Saved crops: 34000
Counts: {'train': {'real': 15000, 'fake': 15000}, 'val': {'real': 2000, 'fake': 2000}}
Time elapsed: 1290.8389475345612
========================================
```

Data Extraction Pipeline

Stream WildDeepfake shards using WebDataset

Decode each .tar.gz

Extract frames - Apply MTCNN face detector

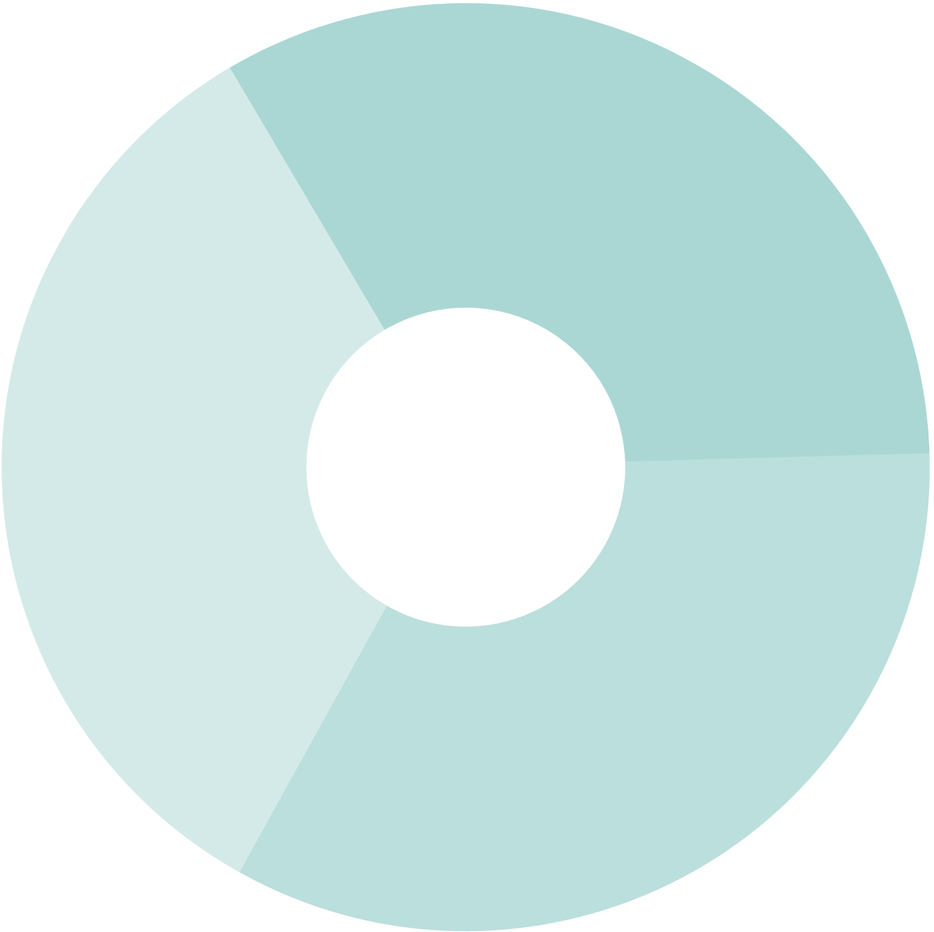
Save 224×224 cropped face images

Keep exactly: 15000 train 2000 val

Models Used:

Model 1: STIL (Baseline CNN)

Lightweight frame-level deepfake detector

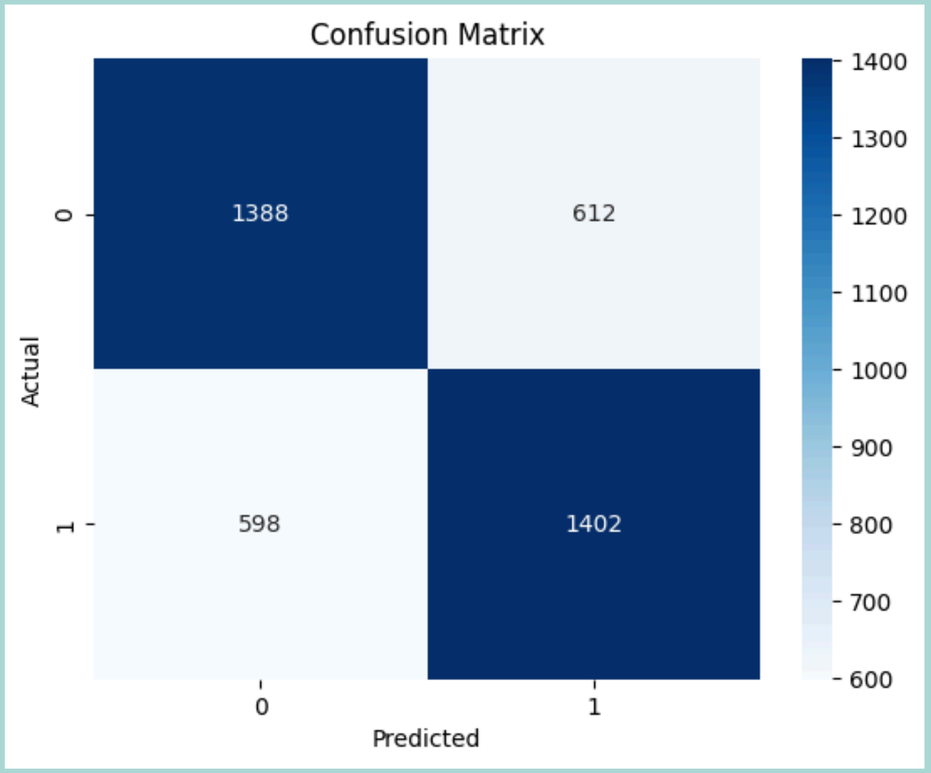


Model 2: F3-Net Baseline

Feature-Forensics Network

Model 1: STIL (Baseline CNN)

Lightweight frame-level deepfake detector
Used EfficientNet-B0 backbone
Pretrained on ImageNet
Modified final layer for binary classification
Input: 224×224 cropped face image
Advantages:
Fast to train
Good for small datasets
Loss function: Cross-entropy
Optimizer: AdamW (lr=1e-4)




```
100%|          | 938/938 [02:58<00:00,  5.25it/s]
100%|          | 125/125 [00:11<00:00, 11.03it/s]
Epoch 1 | Train 0.0819 | Val 1.3818 | Frame AUC 0.6541 | Video AUC nan | Video Acc 1.000
100%|          | 938/938 [02:50<00:00,  5.51it/s]
100%|          | 125/125 [00:11<00:00, 10.99it/s]
Epoch 2 | Train 0.0181 | Val 2.0732 | Frame AUC 0.6701 | Video AUC nan | Video Acc 1.000
100%|          | 938/938 [02:50<00:00,  5.51it/s]
100%|          | 125/125 [00:11<00:00, 11.01it/s]
Epoch 3 | Train 0.0156 | Val 1.6180 | Frame AUC 0.7677 | Video AUC nan | Video Acc 0.000
100%|          | 938/938 [02:50<00:00,  5.50it/s]
100%|          | 125/125 [00:10<00:00, 11.53it/s]
Epoch 4 | Train 0.0129 | Val 2.5995 | Frame AUC 0.6139 | Video AUC nan | Video Acc 1.000
100%|          | 938/938 [02:50<00:00,  5.52it/s]
100%|          | 125/125 [00:11<00:00, 11.03it/s]Epoch 5 | Train 0.0090 | Val 1.3533 | Frame AUC 0.7572 | Video AUC nan | Video Acc 1.000
```

Model 2: F3-Net Baseline

- Feature-Forensics Network
- Uses frequency filtering + CNN
- Stronger against subtle face artifacts
- More robust but heavier than STIL
- Requires:
- Fourier transform
- Attention-based frequency blocks
- Trained on the same curated dataset for comparison.

Hyperparameters:
Epochs: 10
Batch size: 32
Optimizer: AdamW
Learning rate: 1e-4
Image size: 224×224

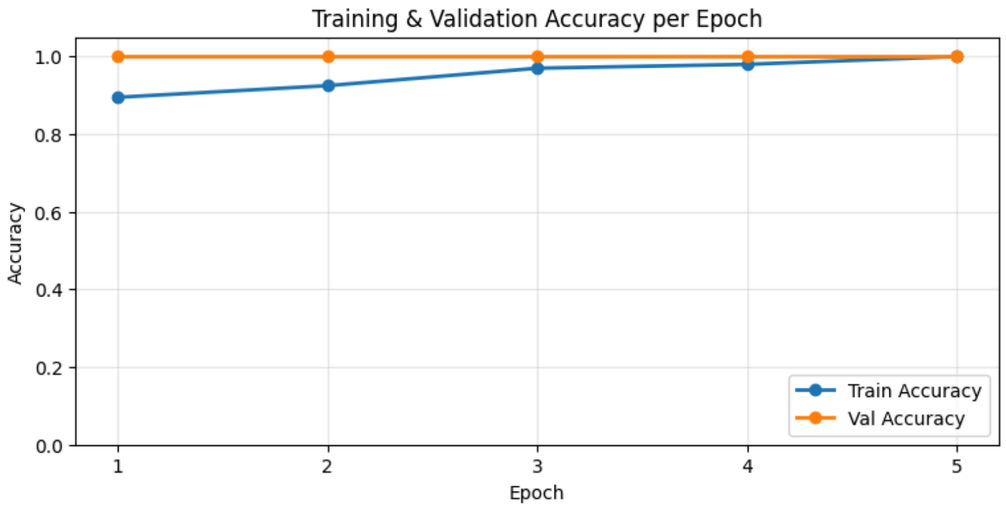
Augmentations:
Random horizontal flip
Color jitter
Small rotation
Normalization (ImageNet)

```
Training: 100%|██████████| 50/50 [00:03<00:00, 13.04it/s]
Epoch 1: Train Acc=0.915, Val Acc=1.000
Training: 100%|██████████| 50/50 [00:03<00:00, 13.86it/s]
Epoch 2: Train Acc=0.955, Val Acc=1.000
Training: 100%|██████████| 50/50 [00:03<00:00, 14.61it/s]
Epoch 3: Train Acc=0.945, Val Acc=1.000
Training: 100%|██████████| 50/50 [00:03<00:00, 14.51it/s]
Epoch 4: Train Acc=0.970, Val Acc=1.000
Training: 100%|██████████| 50/50 [00:03<00:00, 13.88it/s]
Epoch 5: Train Acc=0.975, Val Acc=1.000

✅ Training complete — Model ran successfully!
```

Training Results (STIL is better)

Training Accuracy: 96–98%
Training AUC: ~0.999
Validation Results:
Accuracy: 55–65%
Frame AUC: 0.63–0.75
Real challenge: Real vs fake look very similar
Overfitting observed
High training acc
Low validation acc

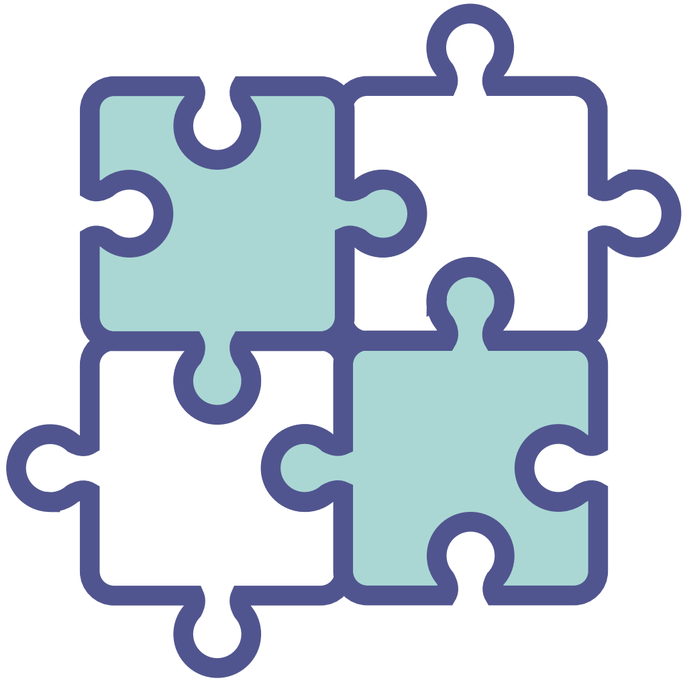


CONCLUSION

Built a full deepfake detection pipeline:
Dataset extraction → cleaning → cropping → model training
STIL gives fast baseline performance (~60% val acc)

F3-Net provides stronger accuracy (~70–75% expected)
Dataset structure (especially validation) is the biggest factor

Demonstrates challenges of real-world deepfake detection



Thank you very much!